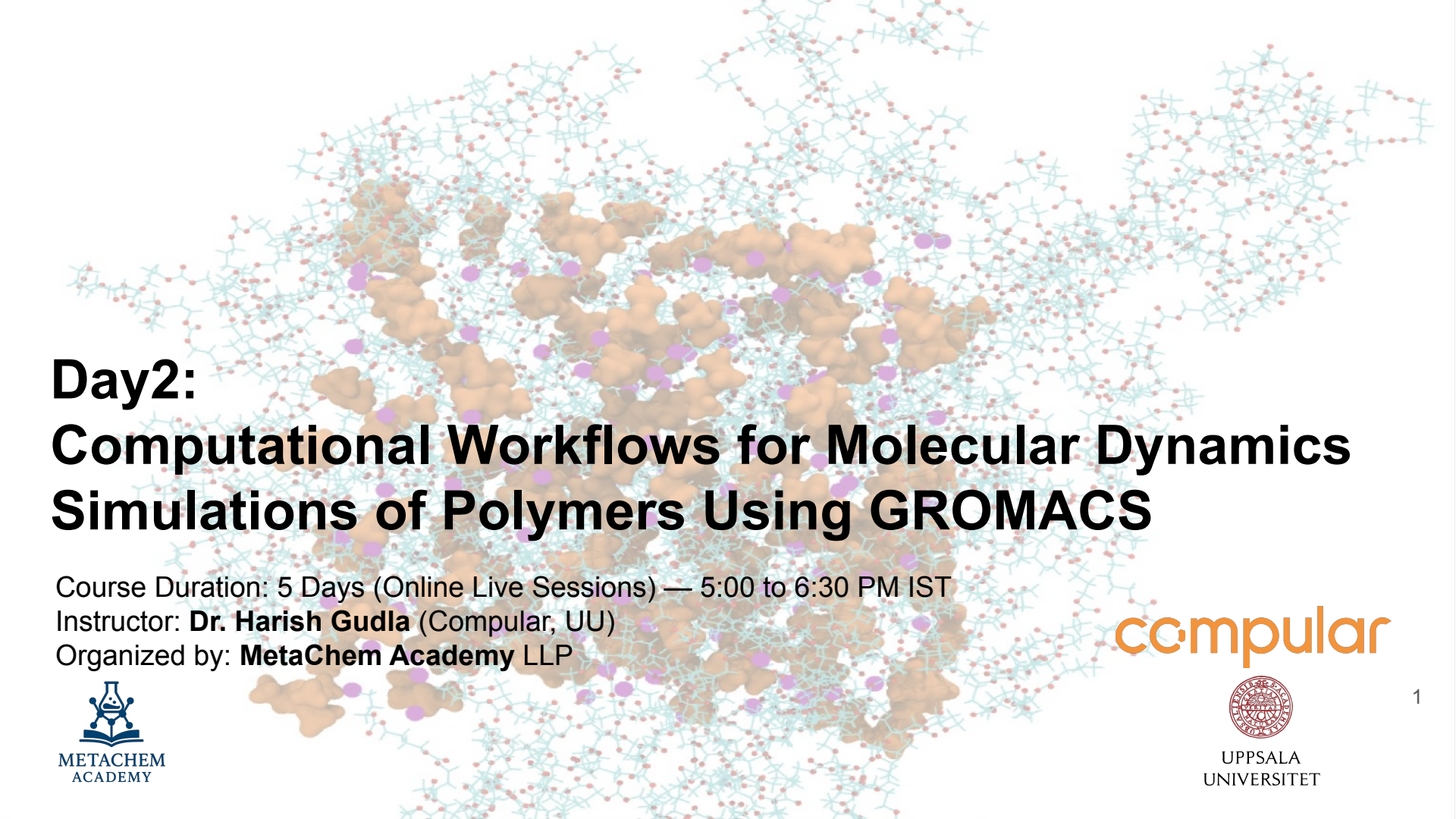




Day2: Computational Workflows for Molecular Dynamics Simulations of Polymers Using GROMACS

Course Duration: 5 Days (Online Live Sessions) — 5:00 to 6:30 PM IST

Instructor: **Dr. Harish Gudla** (Compular, UU)

Organized by: **MetaChem Academy LLP**

Day 2 - Outline

- Recap
- Hands-on session (5:00-6:00 pm IST)
 - Installing the tools (conda environment)
 - Working with python/jupyter notebook
 - Parameterisation of monomer/ligand molecules
- Theory session (6:00-6:30 pm IST)
 - Polymer simulations applications
 - Polymer chain architecture

MD algorithm

Step 1: initial positions $r^{(i=0)}$ and velocities $v^{(i=0)}$
for atoms, set $a = 0$, $t = 0$, $i = 0$, timestep Δt

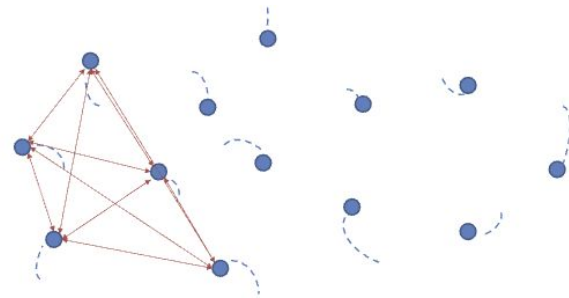
Step 2: after atom movements, location update: $r^p = r^i + v^{(i)}\Delta t + \frac{1}{2}a\Delta t^2 + \text{accurate terms}$, velocity update:
 $v^p = v^{(i)} + a\Delta t + \text{accurate terms}$

Step 3: force/energy calculations: $F = -\nabla v(r^p)$ or $F = F(\Psi(r^p))$ and $a = F / m$

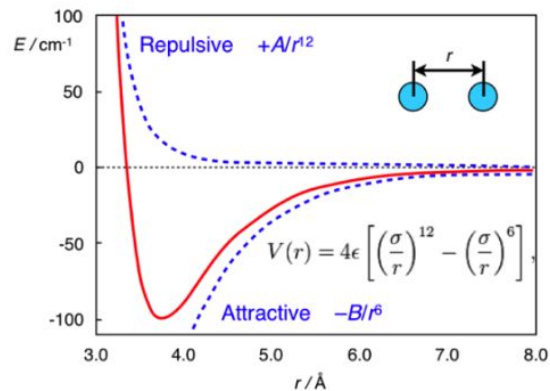
Step 4: after atom movements, location update: $r^{i+1} = r^p + f(a, \Delta t)$, velocity update: $v^{(i+1)} = v^p + f(a, \Delta t)$

Step 5: Apply boundary conditions, temperature and pressure control. Time integration and iteration step: $t = t + \Delta t$, $i = i + 1$. Repeat step 2 until the acceptable results are obtained.

N atoms



Interatomic pair potentials: $U_{1,2}, U_{1,3}, U_{1,4}, U_{1,5}, U_{2,3}, U_{2,4}, U_{2,5}, \dots, U_{N,N-1}$



Classical Force Field

The functional form of a typical forcefield

$$U_{\text{total}} = U_{\text{LJ}} + U_{\text{Coul}} + U_{\text{Bond}} + U_{\text{Angle}} + U_{\text{Dihedral}} + U_{\text{Improper}}$$

Non-bonded interactions

$$U_{\text{LJ}} = 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right],$$

$$U_{\text{Coul}} = \frac{q_i q_j}{4\pi\epsilon_1\epsilon_0 r_{ij}},$$

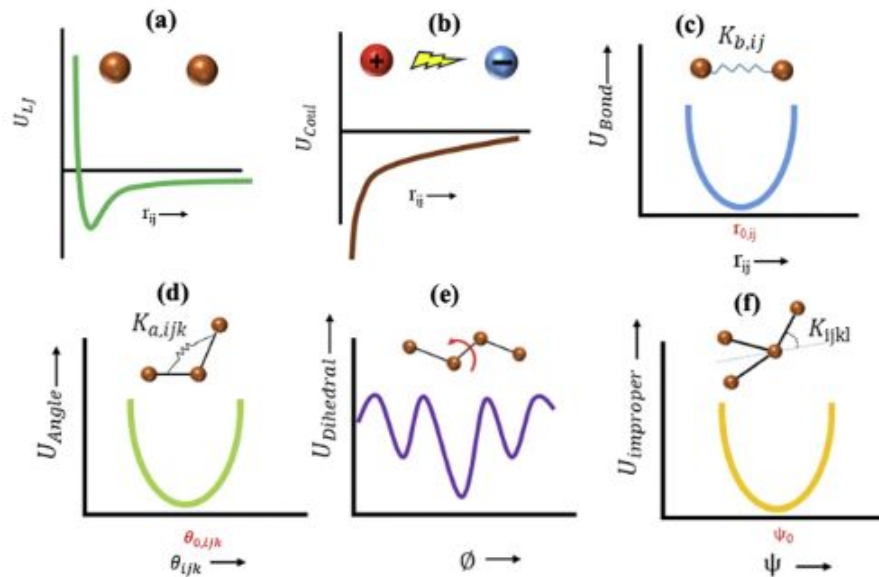
Bonded interactions

$$U_{\text{Bond}} = K_{b,ij}(r_{ij} - r_{0,ij})^2,$$

$$U_{\text{Angle}} = K_{a,ijk}(\theta_{ijk} - \theta_{0,ijk})^2,$$

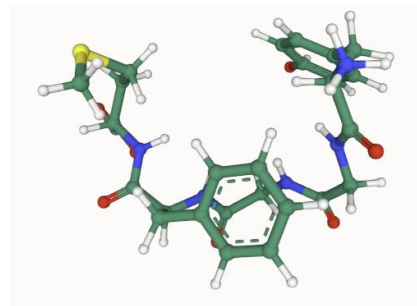
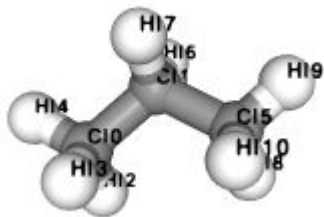
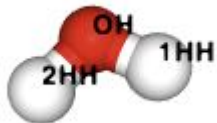
$$U_{\text{Dihedral}} = \frac{1}{2}K_1[1 + \cos(\phi)] + \frac{1}{2}K_2[1 - \cos(2\phi)] \\ + \frac{1}{2}K_3[1 + \cos(3\phi)] + \frac{1}{2}K_4[1 - \cos(4\phi)],$$

$$U_{\text{improper}} = K_{ijkl}(\psi - \psi_0)^2.$$



Classical Force Field

Guess the number of unique parameters for each molecule type



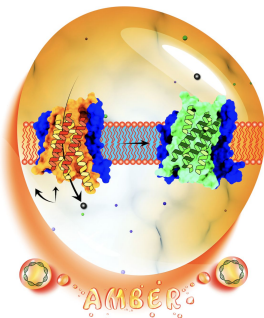
Unique parameters	Water	Propane	Small protein (Met-enkephalin)
Non-bonded	2	6	~30
Partial charges	2	3	~15
Bonds	1	4	~20
Angles	1	6	~50
Dihedrals/Impropers	0	12	~50
Total	6	31	~165

Day 2 - Outline

- Recap
- Hands-on session (5:00-6:00 pm IST)
 - Installing the tools (conda environment)
 - Working with python/jupyter notebook
 - Parameterisation of monomer/ligand molecules
- Theory session (6:00-6:20 pm IST)
 - Polymer simulations applications
 - Polymer chain architecture

Tools and Softwares

- **Python / Jupyter Notebook/Lab** – for workflow automation, running commands, visualising molecules/simulations, execute analysis scripts and plotting the data
- **AmberTools** – Parameterizing monomers using the Generalized Amber Force Field (GAFF) with AmberTools (including antechamber, prepgen, and tleap) to accurately define and construct polymer chains.
- **PACKMOL** – for generating initial multi-chain configurations
- **GROMACS** – performing standard MD equilibration and production simulations, including Energy Minimization (EM), NVT, and NPT simulations. Also, trajectory analysis.

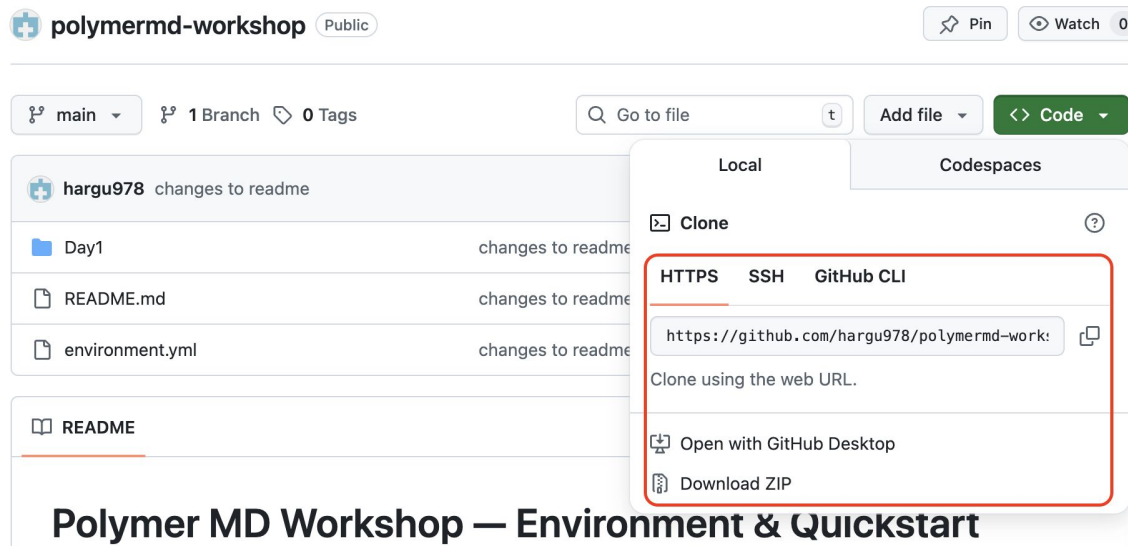


FAST. FLEXIBLE. FREE.
GROMACS

The GROMACS logo, featuring a stylized blue bird in flight.

Installing the tools

- Get this repo: **hargu978/polymermd-workshop**
- If you have git installed:
git clone <https://github.com/hargu978/polymermd-workshop.git>
- Or download as ZIP file
- This contains only materials for today's workshop.
- Follow the steps on installing using **conda**.





127.0.0.1:8888/lab



File Edit View Run Kernel Tabs Settings Help



Name	Modified
/	
Day1	34m ago
conda-setup-guid...	34m ago
conda-setup-guid...	34m ago
environment.yml	34m ago
Install_Wsl_Ubunt...	34m ago
README.md	34m ago
README.pdf	34m ago
setup_conda_env.sh	34m ago

Launcher



Python 3
(ipykernel)



Console



Python 3
(ipykernel)



Other



Terminal



Text File



Markdown File



Python File



Show Conte
Help

Would you like to get
Jupyter news?



Jupyter Notebooks/Labs

- Jupyter Notebook Basics:

<https://jupyter-notebook.readthedocs.io/en/stable/examples/Notebook/Notebook%20Basics.html>

- Jupyter Lab Interface:

<https://jupyterlab.readthedocs.io/en/stable/user/interface.html>

ambertools

1. antechamber

- Generates atom types (GAFF/GAFF2)
- Computes partial charges (AM1-BCC, RESP)
- Produces parameter files (`.mol2`, `.prepi`, `.frcmod`)

2. parmchk2

- Checks for missing parameters
- Generates a `.frcmod` file containing supplemental parameters

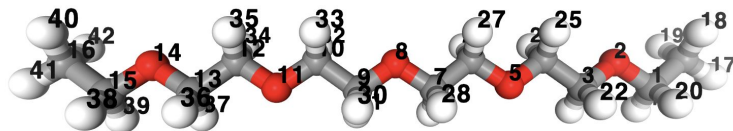
3. tleap

- Assembles the full system (molecule, solvent, ions)
- Writes AMBER `prmtop` and `inpcrd`

4. parmed

- Converts AMBER `prmtop` and `inpcrd` to GROMACS format

Monomer structure



Monomer GAFF atom typing and partial charge parameters

CHARGE	0.00 (0)								
Formula: H26 C12 O5									
ATOM	1	C	PEO	1	0.931	-0.097	-0.023	-0.097100	c3
ATOM	2	C1	PEO	1	2.445	-0.109	0.007	0.132400	c3
ATOM	3	O	PEO	1	2.928	0.653	-1.094	-0.428600	os
ATOM	4	C2	PEO	1	4.355	0.677	-1.114	0.127400	c3
ATOM	5	C3	PEO	1	4.829	1.500	-2.301	0.130400	c3
ATOM	6	O1	PEO	1	6.254	1.515	-2.312	-0.430600	os
ATOM	7	C4	PEO	1	6.750	2.278	-3.409	0.130400	c3
ATOM	8	C5	PEO	1	8.269	2.261	-3.377	0.129400	c3
ATOM	9	O2	PEO	1	8.762	3.024	-4.475	-0.429600	os
ATOM	10	C6	PEO	1	10.186	3.036	-4.489	0.129400	c3
ATOM	11	C7	PEO	1	10.660	3.858	-5.676	0.130400	c3
ATOM	12	O3	PEO	1	12.084	3.872	-5.692	-0.430600	os
ATOM	13	C8	PEO	1	12.575	4.634	-6.792	0.130400	c3
ATOM	14	C9	PEO	1	14.095	4.618	-6.764	0.127400	c3
ATOM	15	O4	PEO	1	14.598	5.379	-7.860	-0.428600	os